Audio file

[6.3 Chronic Leukemia.mp4]

Transcript

Okay, chronic leukemia, that's the next section of this chapter. Before we can understand chronic leukemia, we need to just quickly review normal hematopoiesis. Obviously, we begin with a C34 positive stem cell, as I've stated many times in the prior lectures. That can become a myeloid stem cell or a lymphoid stem cell, and the myeloid stem cell will produce cells of the myeloid lineage, and the lymphoid stem cell will produce cells of the lymphoid lineage. Now, when we talk about a chronic leukemia globally, what we're really referring to is a proliferation of mature lymphocytes. So if there's a neoplastic proliferation in which there are mature lymphocytes floating around in the blood, we call that a chronic leukemia. Again, I've highlighted that here by reminding you that chronic leukemia is a neoplastic proliferation of mature circulating lymphocytes. That's going to result in a high white blood cell count. Now, classically, this disease is insidious in onset. The patients present very slowly and they're often asymptomatic, and they can live a very long time with this disease. Again, it's classically seen in older adults. The prototypic chronic leukemia is chronic lymphocytic leukemia, or CLL. It's a neoplastic proliferation of naive B cells. Naive B cells are the B cells that have just been born from the bone marrow, and one of the characteristic findings is that these cells co-express CD5 and CD20, very important to know for examinations. CD5 is a marker that would normally be expected to be present on a T-cell, but in this particular case, it's aberrantly expressed on this B-cell. And so this is a characteristic finding, co-expression of CD5 and CD20. If you were to look at the blood smear in these patients, there would be increased numbers of lymphocytes with something called SMUD cells, which I'll highlight in a moment. Here is a blood smear showing the increased number of lymphocytes, and in addition, notice that there is this splattered cell appear in the corner, as if someone took their thumb and smashed it or threw it against the slide, and that's called a smud cell. And smud cells are classically seen in CLL. Now, the lymphocytes of CLL can also do what lymphocytes normally do, and that is they can go to the lymph nodes. And so when CLL goes to the lymph node, that will cause generalized lymphadenopathy. At the same time, because the cells now involve the lymph node and are producing a mass, we'll actually call involvement of the lymph node small lymphocytic lymphoma. Again, oma, referring to mass. Some of the complications of CLL are that the patients will develop hypogammaglobulinemia. Now, the cells of CLL are neoplastic B-cells. Normally, a B-cell's goal in life would be to become a plasma cell and secrete immunoglobulin. However, these are neoplastic cells. They have no desire to do anything beneficial, and so they are not going to produce immunoglobulin. And as these cells take over the patient's body, there will be a hypogammaglobulinemia. i.e., a low immunoglobulin level within the blood. And that's important to note because the most common cause of death in these patients is actually infection. Another complication of CLL is autoimmune hemolytic anemia. Not only will these cells not make very much immunoglobulin, but if they happen to try to make immunoglobulin, they do a very bad job. And as a consequence, they actually produce antibody against the patient's own red blood cells. And so that's the way I think about autoimmune hemolytic anemia being a complication of CLL. The third complication that I've listed here is transformation to diffuse large B-cell lymphoma. Patients who have CLL that involves the lymph nodes will obviously have SLL, or small lymphocytic lymphoma, and so they'll have a generalized lymphadenopathy. If the patient-- it is possible for those cells to actually pick up additional mutations and eventually transform to a diffuse large B-cell lymphoma. The classic presentation in this particular case would be that the patient would present with an enlarging lymph node or an enlarging spleen, and the reason that the lymph node would be enlarging would be because they've now got a more aggressive tumor, and that tumor is rapidly growing, enlarging the tissue. So this is an important complication to be aware of as well. Another chronic leukemia would be hairy cell leukemia. This is a neoplastic proliferation of mature B cells, and they're characterized by hairy cytoplasmic processes. One of the very important features of these cells that also helps to make the diagnosis is that the cells are positive for an enzyme called tartrate-resistant acid phosphatase. Here's a classic image of hairy cell leukemia. Again, these are the neoplastic cells, and you can clearly see that these neoplastic cells have these hairy cytoplasmic processes coming off them, and that's one feature of hairy cell leukemia. You can then do a trap stain, tartrate-resistant acid phosphatase, to prove that, indeed, these cells represent hairy cell leukemia. Hairy cell leukemia's clinical features include splenomegaly, the spleen becomes big. Now, this is a very common feature in most chronic leukemias in the sense that when you have a chronic leukemia, the cells are going to go to lymph node and to lymphoid tissues. And so generally speaking, with chronic leukemias, the lymph nodes will become big and the spleen will become big. Normally, however, we would expect the spleen to become big due to expansion of the white pulp, which is where the white blood cells normally live. However, in this case, the splenomegaly is due to expansion of the red pulp, and that is particularly high yield. At the same time, it's important to note that these patients usually lack lymphadenopathy. Finally, if you were to do a bone marrow aspiration on these patients, you would get a dry tap because the bone marrow is often fibrosed in these patients. I sort of put this entire story together by simply remembering the word trap. TRAP, tartrate-resistant acid phosphatase, helps me to remember the chemical reaction that is classic for hairy cell leukemia. It also helps me to remember that the cells get trapped in the red pulp, and they get trapped in the bone marrow. And because they're trapped in the red pulp and trapped in the bone marrow, they can't go to the place where I would normally expect them, which would be in the lymph node. Because I would expect most chronic leukemias would actually cause expansion of the lymph node or enlargement of the lymph node, and here we don't see that. So I use the word trap to really help me remember most of the clinical features of hairy cell leukemia. Very high yield to note that there is an excellent response of hairy cell leukemia to a drug called 2-CDA. The reason why examiners like to go after this is because it is a adenosine deaminase inhibitor. If we inhibit adenosine deaminase, adenosine will accumulate to toxic levels in neoplastic B cells. Recall from biochemistry that adenosine deaminase is actually part of the purine degradation pathway. If we block adenosine deaminase, then obviously adenosine would accumulate, which would then result in death of the neoplastic B cells. Another important chronic leukemia to be familiar with is ATLL, adult T-cell leukemia lymphoma. This is a neoplastic proliferation of mature CD4 positive T-cells, classically associated, very high yield, with HTLV-1, human T-cell leukemia virus 1. And this virus is particularly seen in Japan and in the Caribbean, and that is important to know as well. Clinically, these patients present with rash. I can tell you one general principle of T-cell leukemias is that they like to go to skin and create a rash. Also, the patient will have generalized lymphadenopathy with hepatosplenomegaly. There's nothing special here. We know that mature lymphocytes like to go to lymph node and like to go to spleen, and so we would expect to see lymphadenopathy and hepatosplenomegaly. Finally, what is important and very high yield is that these patients lytic bone lesions with hypercalcemia. The reason why I think this is so high yield is because if I were to tell you that a patient has lytic bone lesions, you would automatically think about maybe something like multiple myeloma. And so for the knee-jerk reaction when we have lytic bone lesions with hypercalcemia, for example, punched out lesions in the bone with hypercalcemia, especially in the discussion of white blood cells, is to think about multiple myeloma. However, patients with adult T-cell leukemia lymphoma can also have lytic bone lesions with hypercalcemia. And in addition, one of the helpful features is that there will also be a rash. So that if on an examination, they tell you that there are lytic bone lesions with hypercalcemia and a rash, take a step back. Don't think about choosing multiple myeloma until you've excluded adult T-cell leukemia lymphoma. Mycosis spongoides is another sort of chronic leukemic process in which the patients get a neoplastic proliferation of mature CD4 positive T-cells. Again, mature CD4 positive T-cells like to go to skin and so the cells will infiltrate the skin and they produce a rash or plaques or nodules, usually multiple throughout the body. If a biopsy was performed, we would see these neoplastic T-cells in the epidermis and it's very important to know that these are called Potrier microabscesses. So if the patient's got accumulation of neoplastic T-cells in the epidermis, we call that a Potrier microabscess. The cells can also spread to the blood. When they spread to the blood, we call that Cesary Syndrome, and the most important thing about Cesary Syndrome is that when we look at the cells under the microscope, they have a classic appearance of cerebroform nuclei. Here's a picture of Cesary Syndrome, and if you look at this nucleus, it sort of looks like there's a bunch of lobes to this nucleus, almost like the brain, because we can see little folding in between those lobes. So you can kind of see this line here, for example, kind of looks like the folds or the creases when you look at the brain. And so they call these cerebroform nuclei. And they are a classic finding in Caesary syndrome. And that concludes the section on chronic leukemia.